VPC-Transit Gateway And Endpoint

Notes: Create a Jira ticket for this assignment, work on the tasks listed below, and submit the challenge completion document by EOD with clear steps and required evidence (outputs/screenshots).

Use Case: Setting up Transit Gateway and VPC Endpoints for a Multi-VPC Architecture

Scenario:

A large organization is migrating its on-premises infrastructure to the AWS cloud.

The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.

The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.

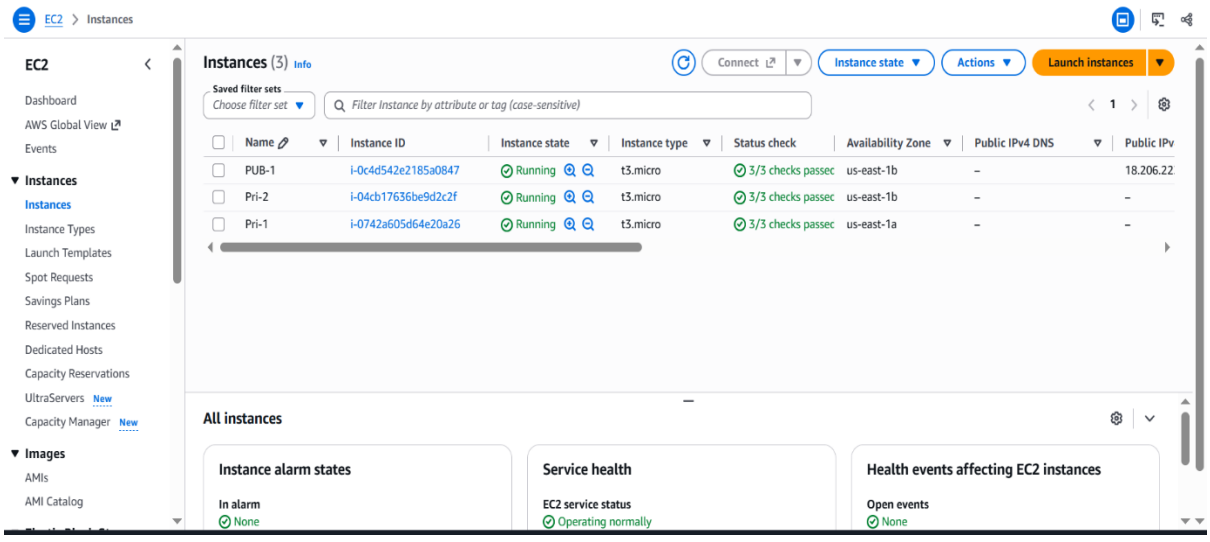
Objectives:

Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.

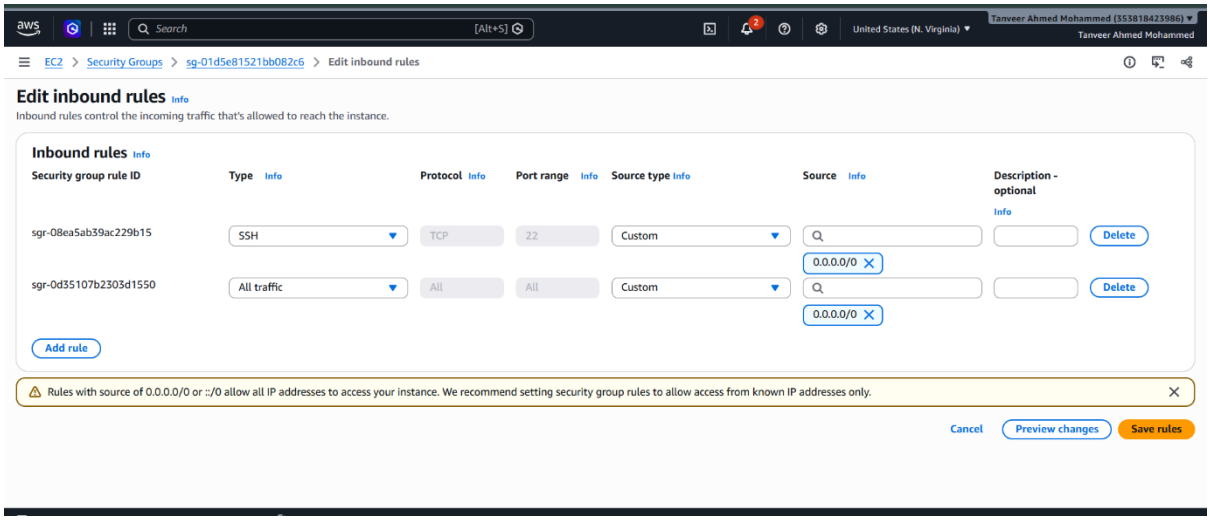
Here I, created a 3 Instance.

1 Public IP

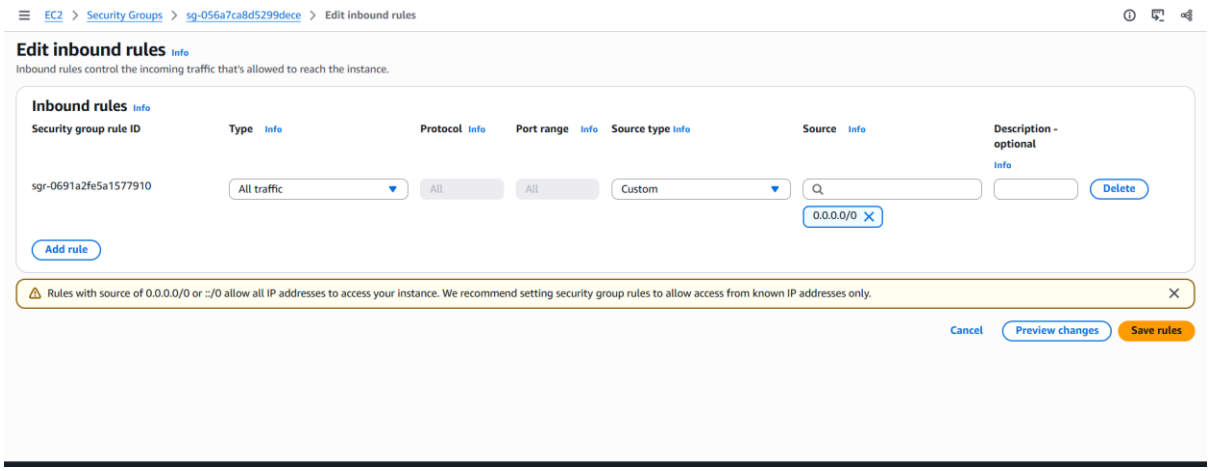
2 Private IP



In Public IP Security Groups

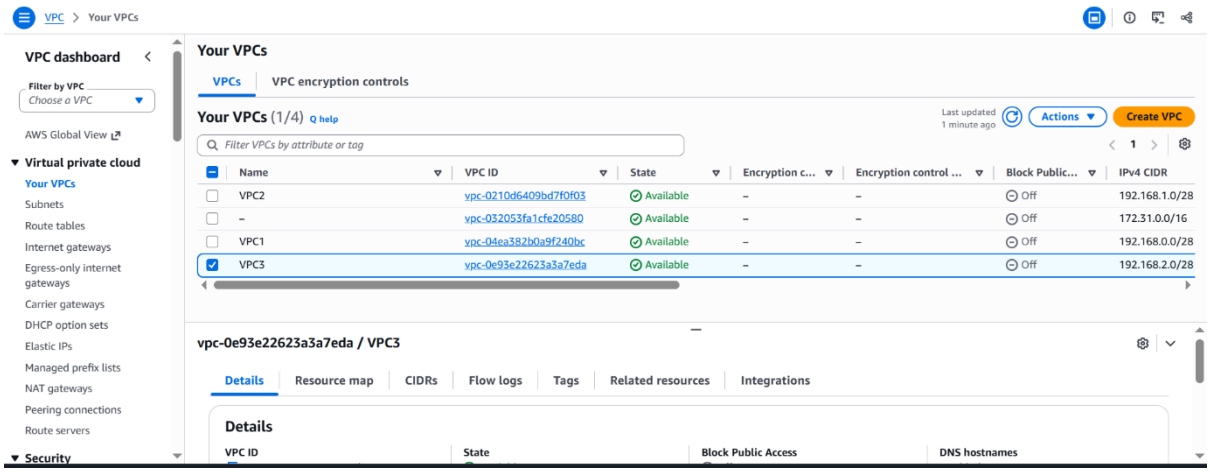


And In Private IP-1 And 2

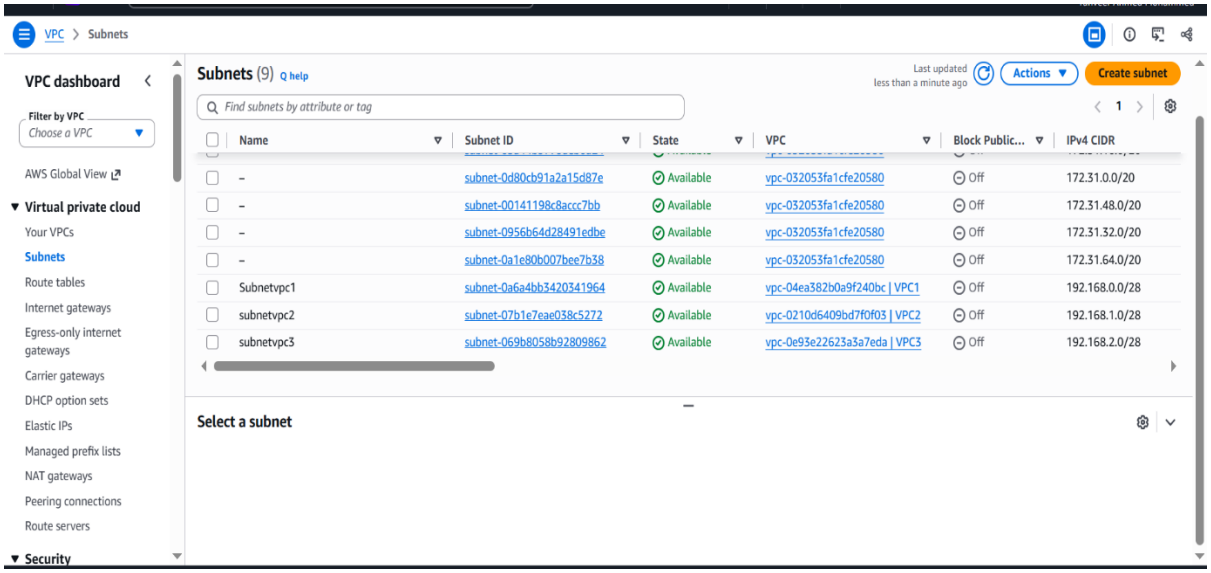


Save And Exit roles.

Next Here Created a 3 VPC’s

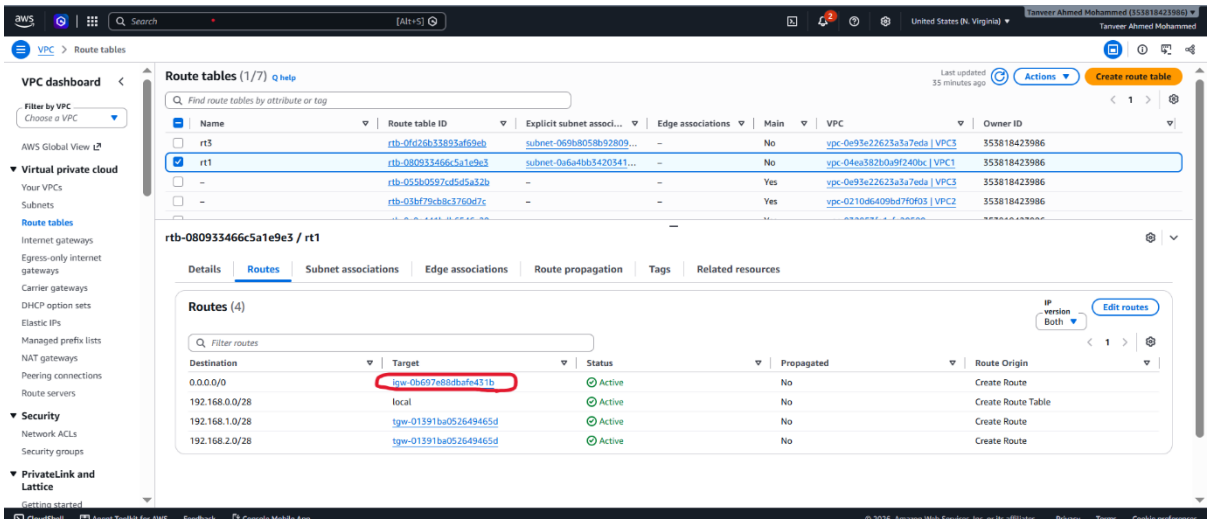


Next, And Subnets

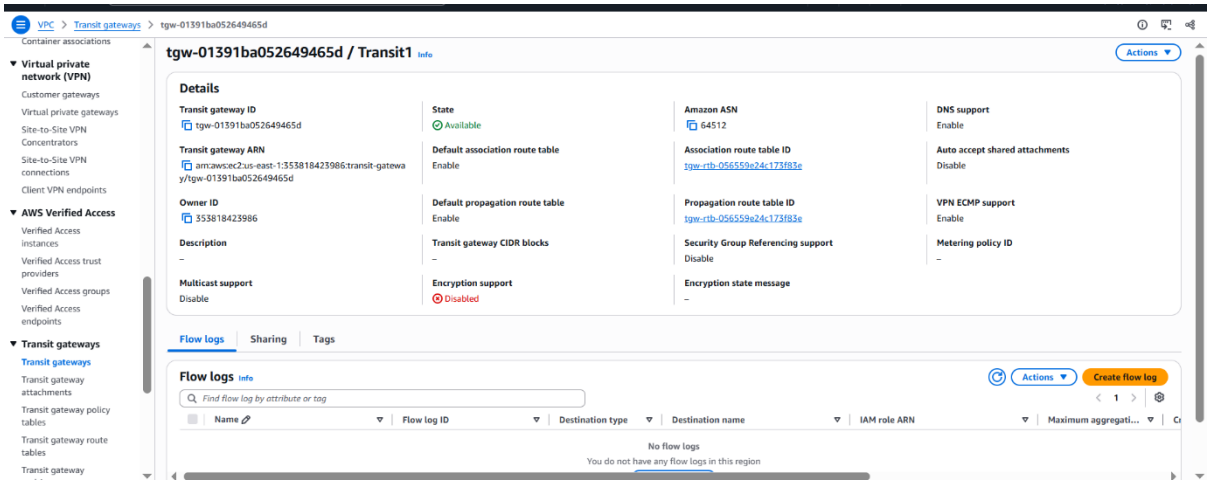


Route Table

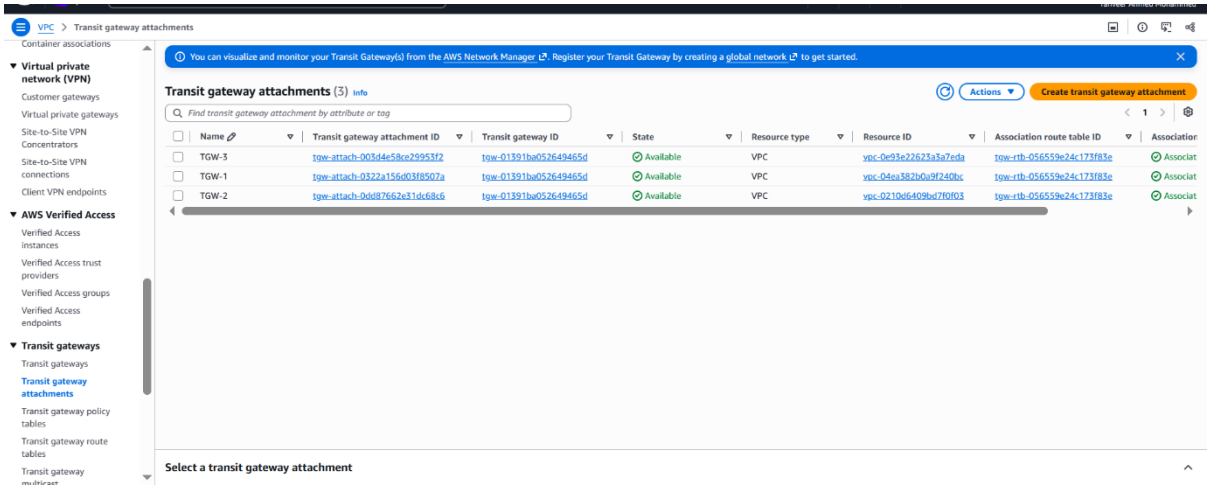
Created 3 routes and on 1st Rt given Internet way for public access.



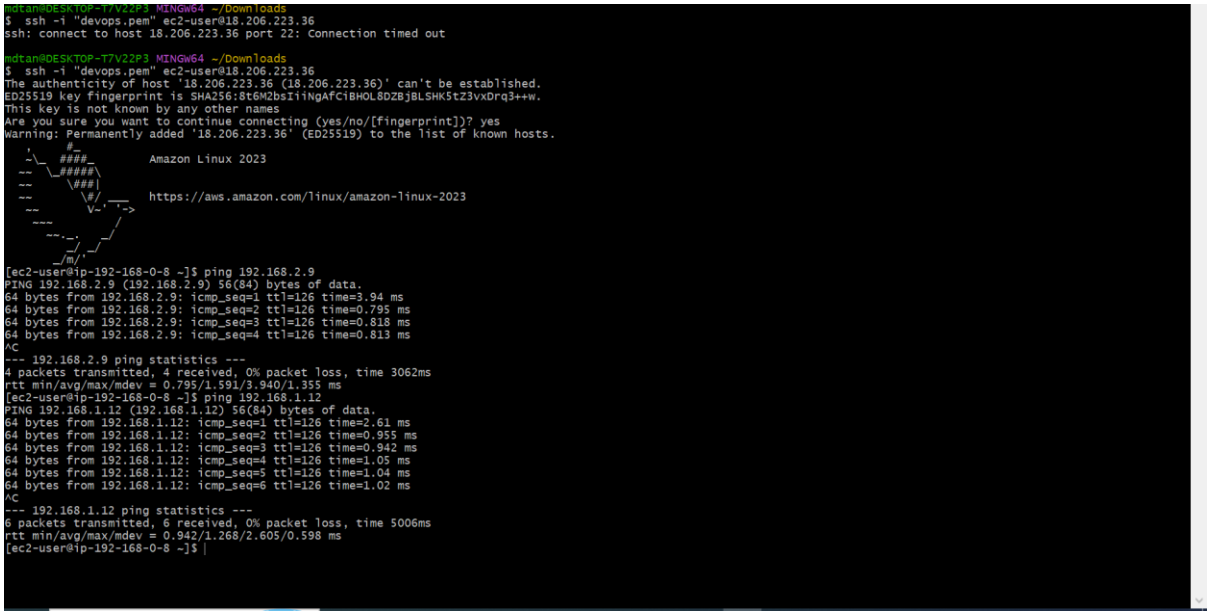
Go, to Next Transit Gateway



Added Transit Attachments



Here Connected with CLI runed with public and private ping is connected to public id address.



Successfully done Transit Gateway merge with three VPC with 1 public and 2 private IP addresses.

Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

Step 1: Setting up Private Key Permissions & Connecting to the Private Instance

- **Navigate to Downloads / Workspace:** You opened your terminal interface (Git Bash / shell) where your .pem key was located.

- **Fix Key Permissions (SSH Key):** If you attempt to use an unprotected private key file, Linux/SSH will reject it with permissions errors. You secured the private key permissions:

Bash

chmod 400 pem.key

(or chmod 400 devops.pem *depending on your key name)*

- **SSH from Public Bastion to Private Instance:** You connected from your jump host/bastion into your private instance inside the VPC using its private IP address:
- Bash

ssh -i pem.key ec2-user@10.0.136.127

(Successfully established connection to Amazon Linux 2023 at IP 10.0.136.127*)*

Step 2: Checking AWS CLI Version & Initial Testing

- **Check AWS CLI Version:** You verified that the AWS Command Line Interface was installed on the instance:

Bash

aws --version

(Output showed: aws-cli/2.38.0 Python/3.9.25 Linux/...*)*

Initial S3 Command Test: You tested listing S3 buckets before configuring credentials:

Bash

aws s3 ls

(Resulted in an error: [ERROR]: An error occurred (NoCredentials): Unable to locate credentials*)*

Step 3: Configuring AWS Credentials

- **Configure Credentials interactively:** You ran the configure command to input your security credentials so the AWS CLI could authenticate API calls:

Bash

aws configure

  - My provided your **AWS Access Key ID** (AKIAV...)
  - My provided your **AWS Secret Access Key** (hvt3J...)
  - Left default region and output format blank.

Step 4: Verifying Configuration Files & Private S3 Endpoint Access

- **Inspect** .aws **Configuration Folder:** You navigated into the hidden AWS directory to verify your configuration and credentials files:

Bash

cd .aws

ls -la

(Listed ., .., config*, and* credentials*)*

View Stored Credentials:

Bash

cat credentials

cat config

(Confirmed your aws_access_key_id *and* aws_secret_access_key *were stored properly under the* [default] *profile)*

Test S3 Access via VPC Endpoint: You ran the S3 listing command again (both with and without explicit profile flags) to verify that data traffic routed privately through your VPC endpoint to Amazon S3:

Bash

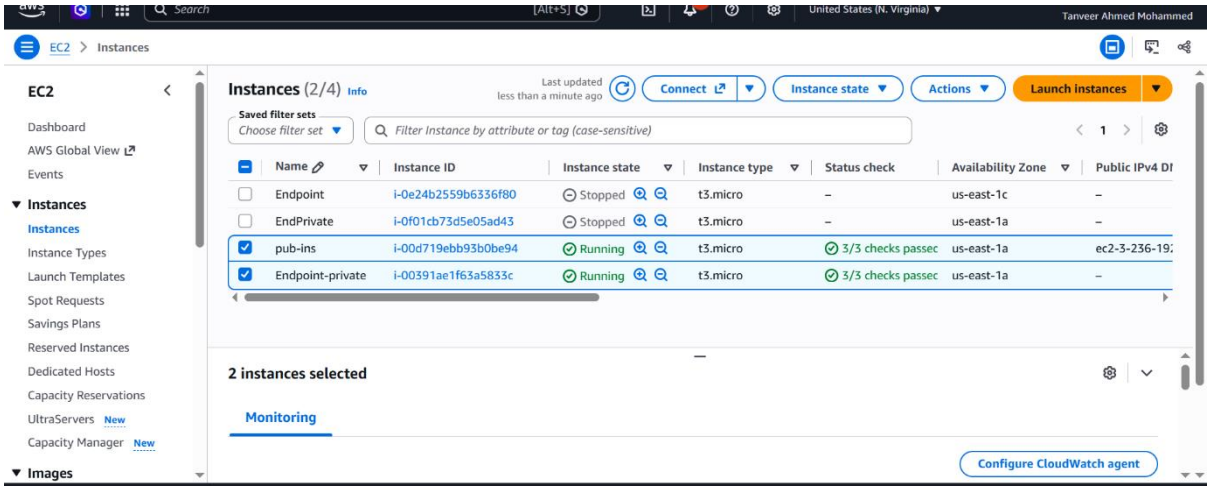
aws s3 ls

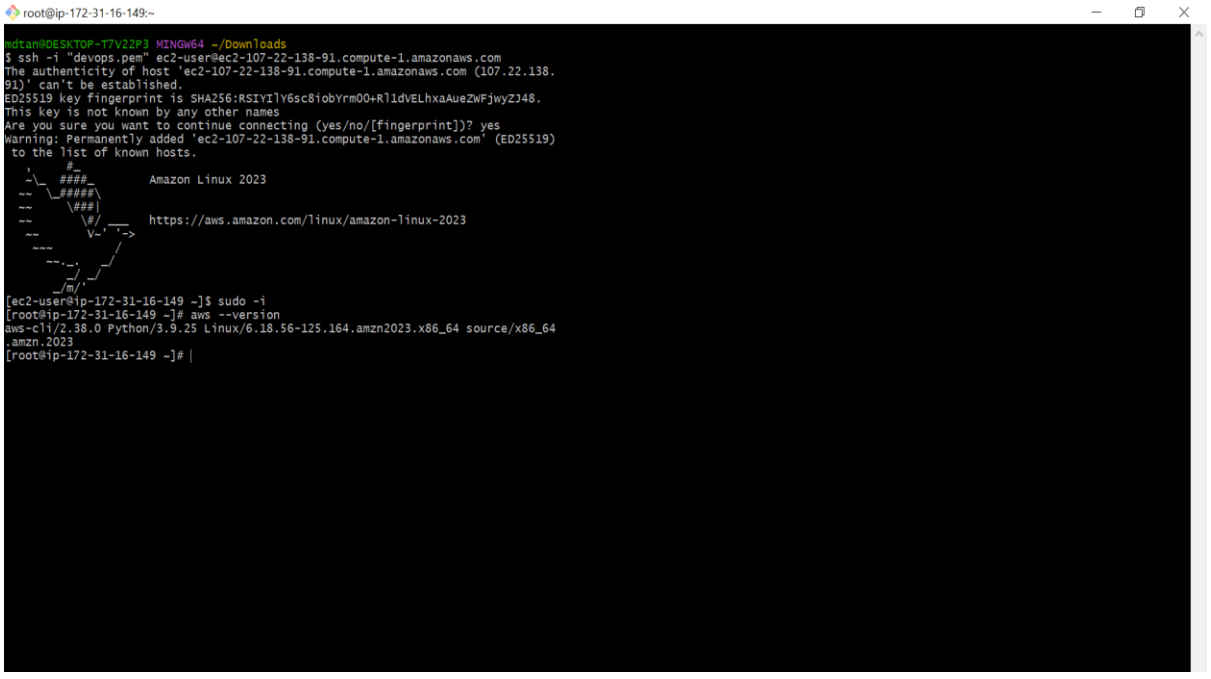
and

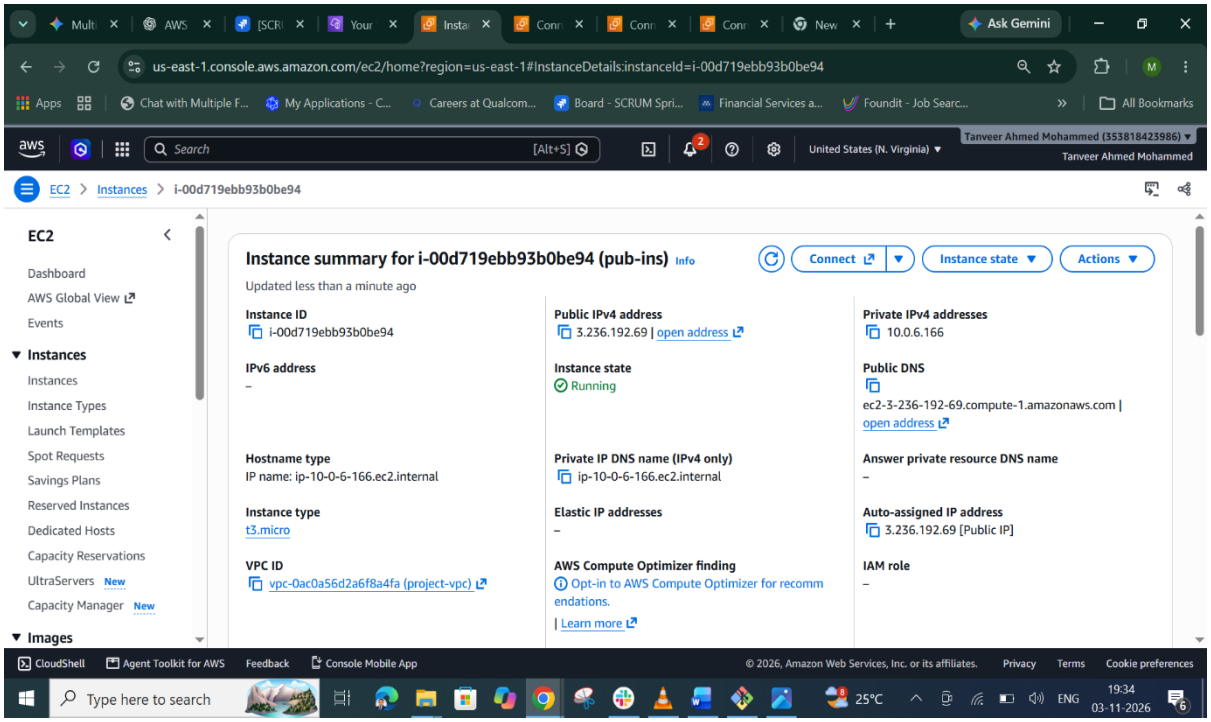
Bash

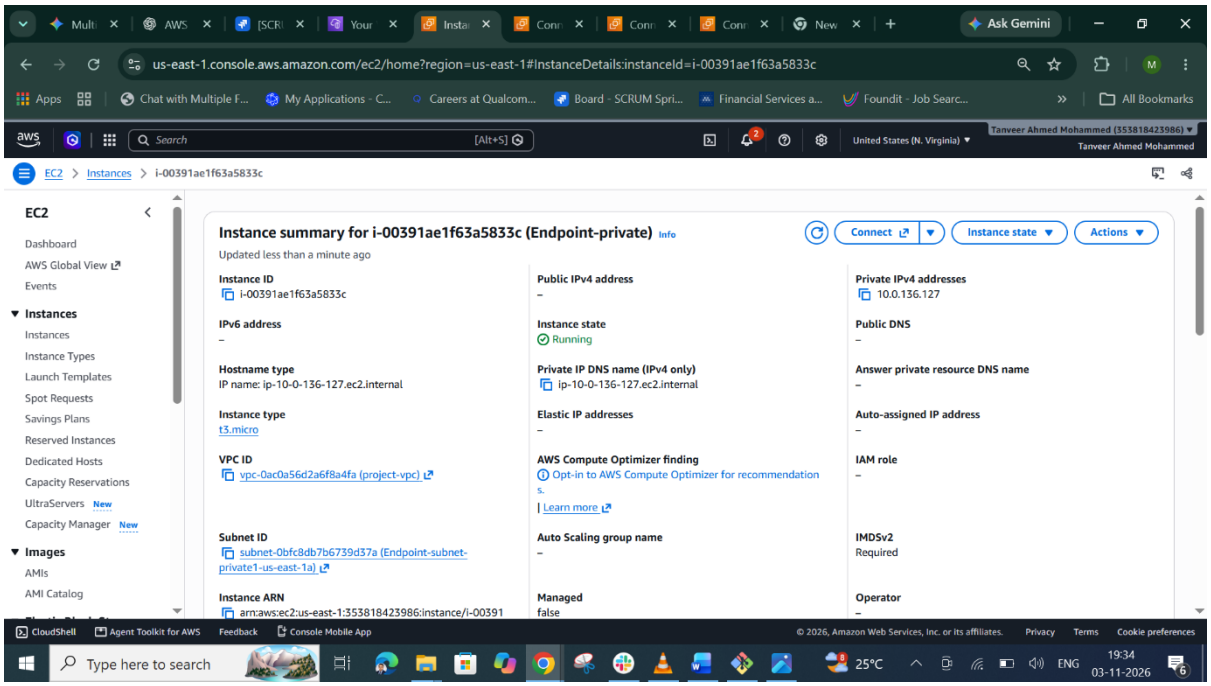
aws s3 ls --profile=default

(Successfully returned your bucket name s3-bucket-pract *without timing out, confirming the private S3 gateway endpoint is functioning correctly)*






us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instanceDetails:instanceId=i-00d719ebb93b0be94
Search [Alt+S]
United States (N. Virginia)
Tanveer Ahmed Mohammed (353818423986)
Tanveer Ahmed Mohammed
EC2 > Instances > i-00d719ebb93b0be94
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
Instance summary for i-00d719ebb93b0be94 (pub-ins) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-00d719ebb93b0be94
IPv6 address
–
Hostname type
IP name: ip-10-0-6-166.ec2.internal
Instance type
t3.micro
VPC ID
vpc-0ac0a56d2a6f8a4fa (project-vpc)
Public IPv4 address
3.236.192.69 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-0-6-166.ec2.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more
Private IPv4 addresses
10.0.6.166
Public DNS
ec2-3-236-192-69.compute-1.amazonaws.com | open address
Answer private resource DNS name
–
Auto-assigned IP address
3.236.192.69 [Public IP]
IAM role
–
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences
Type here to search
25°C
ENG
19:34
03-11-2026


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instanceDetails:instanceId=i-00391ae1f63a5833c
Search [Alt+S]
United States (N. Virginia)
Tanveer Ahmed Mohammed (353818423986)
Tanveer Ahmed Mohammed
EC2 > Instances > i-00391ae1f63a5833c
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Instance summary for i-00391ae1f63a5833c (Endpoint-private) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-00391ae1f63a5833c
IPv6 address
–
Hostname type
IP name: ip-10-0-136-127.ec2.internal
Instance type
t3.micro
VPC ID
vpc-0ac0a56d2a6f8a4fa (project-vpc)
Subnet ID
subnet-0bfc8db7b6739d37a (Endpoint-subnet-private1-us-east-1a)
Instance ARN
arn:aws:ec2:us-east-1:353818423986:instance/i-00391
Public IPv4 address
–
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-0-136-127.ec2.internal
Elastic IP addresses
–
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more
Auto Scaling group name
–
Managed
false
Private IPv4 addresses
10.0.136.127
Public DNS
–
Answer private resource DNS name
–
Auto-assigned IP address
–
IAM role
–
IMDSv2
Required
Operator
–
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Privacy
Terms
Cookie preferences
Type here to search
25°C
ENG
19:34
03-11-2026

ec2-user@ip-10-0-136-127:~

```
mdtan@DESKTOP-T7V22P3 MINGW64 ~/Downloads
$ ssh -i "devops.pem" ec2-user@10.0.136.127


mdtan@DESKTOP-T7V22P3 MINGW64 ~/Downloads
$ ssh -i "devops.pem" ec2-user@10.0.7.163
ssh: connect to host 10.0.7.163 port 22: Connection timed out

mdtan@DESKTOP-T7V22P3 MINGW64 ~/Downloads
$ ssh -i "devops.pem" ec2-user@3.236.192.69
The authenticity of host '3.236.192.69 (3.236.192.69)' can't be established.
ED25519 key fingerprint is SHA256:f+CSMM+uavKWRamK7DsgU0j1Dk1jro9vL6jGBwV9hoA.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.236.192.69' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-6-166 ~]$ sudo -i
[root@ip-10-0-6-166 ~]# nano pem.key
[root@ip-10-0-6-166 ~]# chmod 400
chmod: missing operand after '400'
Try 'chmod --help' for more information.
[root@ip-10-0-6-166 ~]# chmod 400 pem.key
[root@ip-10-0-6-166 ~]# ssh -i pem.key ec2-user@10.0.136.127
The authenticity of host '10.0.136.127 (10.0.136.127)' can't be established.
ED25519 key fingerprint is SHA256:OXJfRSWBpMNG4mXTRTDf7BiH8lSIpXX86CwKwyvVvfU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.136.127' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
```

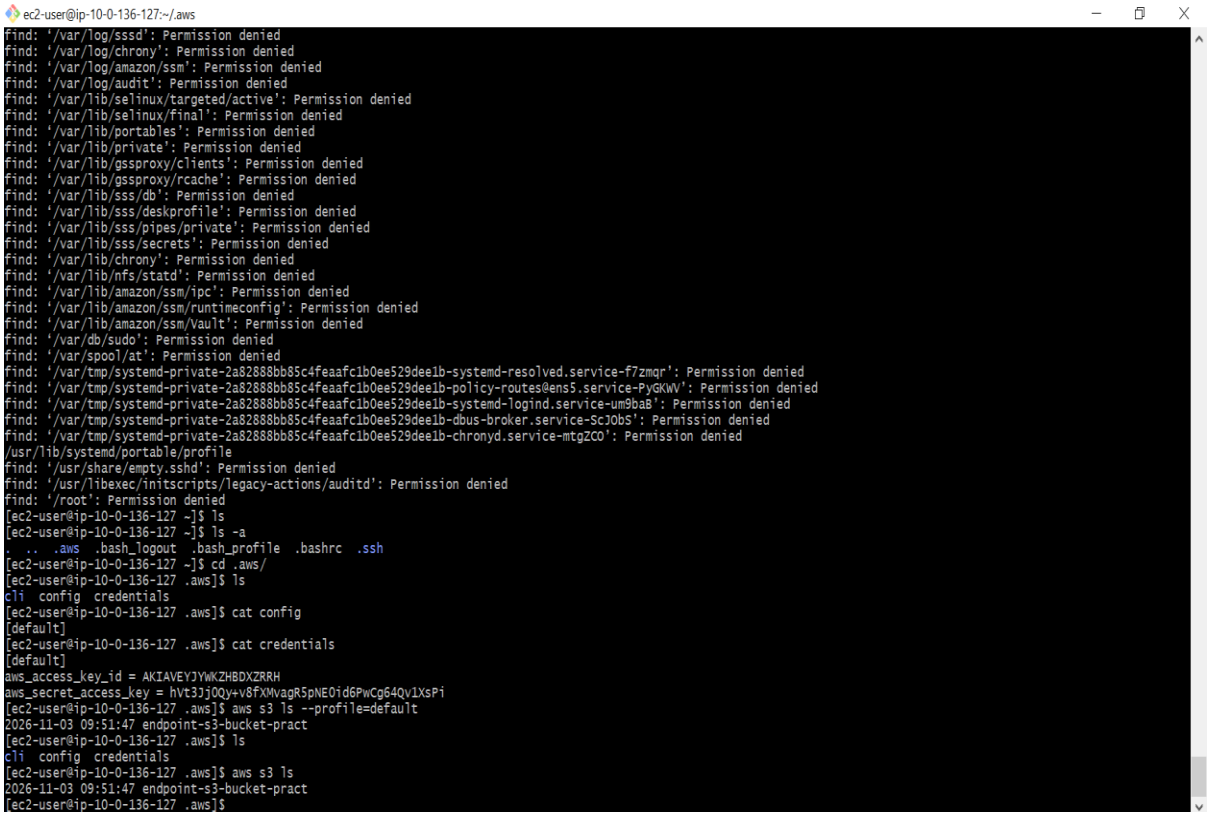
ec2-user@ip-10-0-136-127:~

```
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-6-166 ~]$ sudo -i
[root@ip-10-0-6-166 ~]# nano pem.key
[root@ip-10-0-6-166 ~]# chmod 400
chmod: missing operand after '400'
Try 'chmod --help' for more information.
[root@ip-10-0-6-166 ~]# chmod 400 pem.key
[root@ip-10-0-6-166 ~]# ssh -i pem.key ec2-user@10.0.136.127
The authenticity of host '10.0.136.127 (10.0.136.127)' can't be established.
ED25519 key fingerprint is SHA256:OXJfRSWBpMNG4mXTRTDf7BiH8lSIpXX86CwKwyvVvfU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.136.127' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-136-127 ~]$ aws --version
aws-cli/2.38.0 Python/3.9.25 Linux/6.18.56-125.164.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-10-0-136-127 ~]$ aws s3 ls

aws: [ERROR]: An error occurred (NoCredentials): Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-10-0-136-127 ~]$ aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

AWS Access Key ID [None]: AKIAVEYJYWKZHBDXZRRH
AWS Secret Access Key [None]: hVt3JjOQy+v8fXMvagR5pNEOid6PwCg64Qv1XsPi
Default region name [None]:
Default output format [None]: [ec2-user@ip-10-0-136-127 ~]$
```



(Here,Successfully returned your bucket name s3-bucket-pract *without timing out, confirming the private S3 gateway endpoint is functioning correctly)*